Shang Yang

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Education

Massachusetts Institute of Technology (MIT) Sep. 2023 - Present
Ph.D. Student in EECS, advised by Prof. Song Han. Cambridge, MA

Tsinghua University Aug. 2019 - Jul. 2023
Bachelor of Engineering in Electronic Information Science and Technology Beijing, China

- Overall GPA: 3.99 / 4.0 Rank: 1 / 256

Research Interests

Machine Learning Systems; Efficient LLM Post-Training; LLM Agents & Multi-Agent Systems; Model Compression, Quantization & Sparsity; Efficient Inference & Serving.

Selected Publications Google Scholar

* Equal contribution · † System lead · **6,000+ total citations** as of Nov. 2026.

[1] Ji Lin*, Jiaming Tang*, Haotian Tang*†, **Shang Yang***†, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, Song Han. ***AWQ: Activation-aware Weight Quantization for LLM Compression and Acceleration.*** **(MLSys 2024 Best Paper Award)**

[2] **Shang Yang**, Zhichen Zeng, Shi Dong, Yichuan Wang, Siqi Chen, Junxian Guo, Joseph E. Gonzalez, Banghua Zhu, Song Han. ***OARS: On Post-Training for Cost-Efficient Heterogeneous Multi-Agent Systems.*** **(Preprint, 2026)**

[3] Qinghao Hu*, **Shang Yang***, Junxian Guo, Xiaozhe Yao, Yujun Lin, Yuxian Gu, Han Cai, Chuang Gan, Ana Klimovic, Song Han. ***Taming the Long-Tail: Efficient Reasoning RL Training with Adaptive Drafter.*** **(ASPLOS 2026, MIT Homepage)**

[4] **Shang Yang***, Junxian Guo*, Haotian Tang, Qinghao Hu, Guangxuan Xiao, Jiaming Tang, Yujun Lin, Zhijian Liu, Yao Lu, Song Han. ***LServe: Efficient Long-sequence LLM Serving with Unified Sparse Attention.*** **(MLSys 2025)**

[5] Yujun Lin*, Haotian Tang*, **Shang Yang***, Zhekai Zhang, Guangxuan Xiao, Chuang Gan, Song Han. ***QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving.*** **(MLSys 2025)**

[6] Zhuoyang Zhang*, **Shang Yang***, Qinghao Hu, Luke J. Huang, James Hou, Yufei Sun, Yao Lu, Song Han. ***ForeAct: Steering Your VLA with Efficient Visual Foresight Planning.*** **(CVPR 2026 Highlight)**

[7] Haotian Tang*, **Shang Yang***, Zhijian Liu, Ke Hong, Zhongming Yu, Xiuyu Li, Guohao Dai, Yu Wang, Song Han. ***TorchSparse++: Efficient Training and Inference Framework for Sparse Convolution on GPUs.*** **(MICRO 2023)**

[8] Luke J. Huang, Zhuoyang Zhang, Qinghao Hu, **Shang Yang**, Song Han. ***Stable Asynchrony: Variance-Controlled Off-Policy RL for LLMs.*** **(ICML 2026)**

[9] Ruitao Liu*, Qinghao Hu*, Alex Hu, Yecheng Wu, **Shang Yang**, Luke J. Huang, Zhuoyang Zhang, Han Cai, Song Han. ***Hide to Guide: Learning via Semantic Masking.*** **(NeurIPS 2026)**

[10] Zhuoyang Zhang*, Luke J. Huang*, Chengyue Wu, **Shang Yang**, Yao Lu, Song Han. ***Locality-aware Parallel Decoding for Efficient Autoregressive Image Generation.*** **(ICLR 2026 Oral)**

[11] Jiaming Tang*, Yufei Sun*, Yilong Zhao, **Shang Yang**, Yujun Lin, Zhuoyang Zhang, James Hou, Yao Lu, Zhijian Liu, Song Han. ***VLASH: Real-Time VLAs via Future-State-Aware Asynchronous Inference.*** **(IROS 2026)**

[12] Yuxian Gu, Qinghao Hu, **Shang Yang**, Haocheng Xi, Junyu Chen, Song Han, Han Cai. ***Jet-Nemotron: Efficient Language Model with Post Neural Architecture Search.*** **(NeurIPS 2025)**

[13] Haotian Tang*, Yecheng Wu*, **Shang Yang**, Enze Xie, Junsong Chen, Junyu Chen, Zhuoyang Zhang, Han Cai, Yao Lu, Song Han. ***HART: Efficient Visual Generation with Hybrid Autoregressive Transformer.*** **(ICLR 2025)**

[14] Guangxuan Xiao, Jiaming Tang, Jingwei Zuo, Junxian Guo, **Shang Yang**, Haotian Tang, Yao Fu, Song Han. ***DuoAttention: Efficient Long-Context LLM Inference with Retrieval and Streaming Heads.*** **(ICLR 2025)**

[15] Junyu Chen*, Han Cai*, Junsong Chen, Enze Xie, **Shang Yang**, Haotian Tang, Muyang Li, Yao Lu, Song Han. ***Deep Compression Autoencoder for Efficient High-Resolution Diffusion Models.*** **(ICLR 2025)**

Selected Open-Source Projects

mit-han-lab/llm-awq (3.7K Stars)

- Low-bit LLM quantization and efficient GPU inference with AWQ and TinyChat.
- Adopted by NVIDIA TensorRT-LLM, Hugging Face Transformers, vLLM, SGLang, etc. AWQ-quantized models have **200M+ downloads** on Hugging Face.
- System lead: quantization, GPU kernels, inference runtime, and deployment.

NVlabs/VILA (3.9K Stars)

- Efficient vision-language models for image and video understanding.
- Adopted by NVIDIA Cosmos, NIM, Jetson Platform Services, and the VSS Blueprint. The VILA model family has **5M+ downloads** on Hugging Face.
- Inference lead and core contributor: quantization, GPU kernels, runtime, and performance.

mit-han-lab/omniserve (800+ Stars)

- Unified LLM serving with low-bit quantization (QServe) and sparse attention (LServe).
- Quantization adopted by TensorRT-LLM and SGLang. Up to **3.5**× serving throughput vs. TensorRT-LLM.
- Project lead: quantization and sparsity algorithms, GPU kernels, serving runtime, and integration.

mit-han-lab/torchsparse (1.5K Stars)

- Efficient GPU sparse-convolution library for training and inference in 3D deep learning.
- Adopted by Microsoft TRELLIS, Waabi Flux4D, One-2-3-45, Argoverse 2, etc.
- Project lead: algorithm and system architecture, GPU kernels, runtime, and deployment.

mit-han-lab/fastrl

- Efficient RL framework for reasoning LLM post-training.
- Up to **2.1**× end-to-end RL training speedup over verl, with lossless on-policy rollouts.
- Project lead: algorithm design, system architecture, and RL training framework.

Teaching and Service

Co-Instructor & Lecturer, MIT 6.5940: TinyML and Efficient AI Computing (∼400 students) Fall 2026

Teaching Assistant, MIT 6.5940: TinyML and Efficient Deep Learning Computing Fall 2024

Conference Reviewer: MLSys, NeurIPS, ICML, ICLR, CVPR, ICCV, AAAI, etc.

Invited Talks

[12/2025] Physical Intelligence – ForeAct: Steering Your VLA with Efficient Visual Foresight Planning

[06/2025] Meta – LServe: Efficient Long-sequence LLM Serving with Unified Sparse Attention

[05/2025] MLSys – QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving

[05/2025] MLSys – LServe: Efficient Long-sequence LLM Serving with Unified Sparse Attention

[04/2025] Databricks – QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving

[05/2024] NVIDIA – QServe: W4A8KV4 Quantization and System Co-design for Efficient LLM Serving

[10/2023] MICRO – TorchSparse++: Efficient Training and Inference Framework for Sparse Convolution